

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

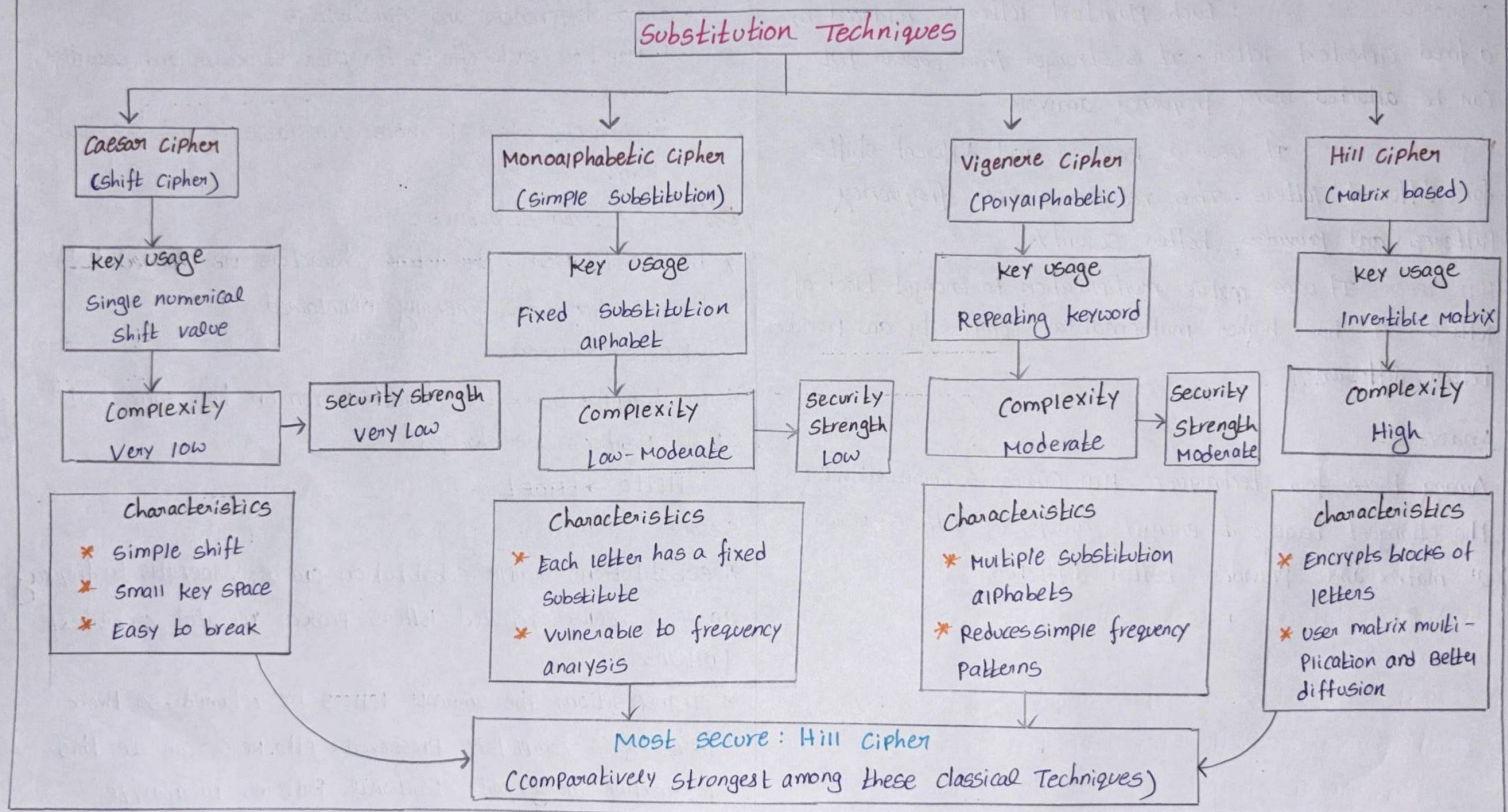
CONCEPT MAPPING

Code of Conduct:

I Ch. Nisha (192324306) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ch. Nisha

8, Develop a concept map comparing substitution techniques: Caesar cipher, Monoalphabetic cipher, Vigenere cipher and Hill cipher. Perform the following tasks to connect based on: Key usage, Complexity and security strength. Analyze which technique is most secure and why.



Explanation:

Caesar cipher: Each letter is shifted by a fixed number. It has a very small key space and can easily be broken using brute force.

Monoalphabetic cipher: Each plaintext letter is replaced by a fixed ciphertext letter. It is stronger than Caesar but can be attacked using frequency analysis.

Vigenere cipher: It uses a keyword and different shifts for different letters. This reduces simple frequency patterns and provides better security.

Hill cipher: It uses matrix multiplication to encrypt block of letters. It has higher mathematical complexity and provides better diffusion.

Analysis:

Among these four techniques, Hill cipher is comparatively the strongest because it encrypts groups of letters using a matrix and provides better diffusion.

3) construct a concept map comparing substitution techniques, classical encryption schemes: substitution techniques and Transposition Techniques. perform the following task:

- a. show differences and similarities
 - i. Map how each affects the data structure and security level.
 - ii. Analyze which is more vulnerable to attacks and why.

Effect on data structure:

* In substitution, the actual characters are changed but their positions are generally maintained.

HELLO → KHOOR

* In transposition, the characters remain the same but their positions are changed.

HELLO → LHOEL

Security Analysis:

* substitution: simple substitution can be vulnerable to frequency analysis because repeated letters produce repeated ciphertext patterns.

* Transposition: The original letters are retained, so their frequency is completely preserved. Attackers can use this information along with positional patterns to analyze

Classical Encryption

(Early methods used to secure information)

Substitution Techniques

(Replace plaintext symbols with other symbols)

Operation → Replaces characters with other characters

Effect on Data structure → changes the actual symbol values.

Examples → Caesar cipher, Monoalphabetic cipher, Vigenere cipher, Hill Cipher.

Security → vulnerable to frequency analysis. security depends on substitution

Similarities

- * Both are classical methods
- * Both use a key or rule
- * Both convert plaintext to ciphertext

Difference

- * Substitution changes symbols
- * Transposition rearranges symbols.
- * Data values change vs remain same

Transposition Techniques

(Rearrange the position of symbols)

Operation → rearranges the positions of characters

Effect on Data structure → characters remain the same but their positions are changed

Examples → Rail Fence cipher, columnar Transposition, Double Transposition

Security → Frequency of letters is preserved. vulnerable to structural and pattern analysis.

Both are vulnerable to cryptanalysis;
Combining substitution + transposition improves complexity